Python for Finance

Companion Slides to the Practical Sessions (TD 1–6)

M1 Economie – Finance, Université Paris Dauphine

Matéo Molinaro

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How to Use These Slides

These slides are a **companion** to the Jupyter notebooks used in the practical sessions (TD). They give the **full derivations** behind each practical session, self-contained, so that the slide deck can be read on its own. They point back to the notebook exercise that implements each result – but the derivations themselves do not depend on the code.

Companion sessions

TD1: Python foundations, Black–Scholes Greeks, Monte Carlo • TD2: Gradient descent, logistic MLE • TD3: Time series (AR/ARIMA/GARCH), OLS • TD4: VaR, backtesting, CoVaR • TD5: Markowitz optimization • TD6: Index construction

Session Map

Session	Topic	Notebook
TD 1	Python foundations, BSM Greeks, Monte Carlo	*Exercices - M1_eco_fi.ipynb*
TD 2	Gradient descent, logistic MLE	*Gradient Descent Students.ipynb*
TD 3	Time series: AR/ARIMA/GARCH, OLS	*Session 3 Time series modelling.ipynb*
TD 4	VaR, backtesting, CoVaR	*Session 4 Systemic risk ... .ipynb*
TD 5	Markowitz optimization	*TP Markowitz CAC 40.ipynb*
TD 6	Index construction	*Index_construction_introduction.ipynb*

Roadmap

1. TD4 – Value at Risk, Backtesting & CoVaR

Outline

1. TD4 – Value at Risk, Backtesting & CoVaR

The Quantile Function

Let X be a random variable with CDF

$$\boxed{F_X(x) = \mathbb{P}(X \leq x)}.$$

For a probability level $\alpha \in (0, 1)$, the α-**quantile** of X is

$$\boxed{q_\alpha(X) = F_X^{-1}(\alpha) = \inf\{x \in \mathbb{R} : F_X(x) \geq \alpha\}}.$$

In words, q_α is the value below which approximately $100\alpha\%$ of the distribution lies. For a continuous, strictly increasing CDF,

$$\boxed{F_X(q_\alpha) = \alpha} \qquad \Longleftrightarrow \qquad \boxed{\mathbb{P}(X \leq q_\alpha) = \alpha}.$$

Example

If $q_{0.05}(R) = -0.0225$, then $\mathbb{P}(R \leq -0.0225) = 0.05$: approximately 5% of returns lie below -2.25%.

The Empirical Quantile

Given a sample $X_1, \ldots, X_T$, sort the observations:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(T)}.$$

A simple empirical estimator of the α-quantile is

$$\boxed{\widehat{q}_\alpha = X_{(\lceil \alpha T \rceil)}}.$$

Remark

Statistical software may interpolate between adjacent order statistics, so the exact sample quantile can depend slightly on the interpolation convention used.

Definition of Value at Risk

Let R_{t+1} denote the portfolio return over the risk horizon. If VaR is reported as a **positive loss number**, the VaR at confidence level $1-\alpha$ is

$$\boxed{\mathrm{VaR}_{1-\alpha} = -q_\alpha(R_{t+1})} \quad \Longleftrightarrow \quad \boxed{\mathrm{VaR}_{1-\alpha} = -F_R^{-1}(\alpha)}.$$

Example: 95% VaR

$\alpha = 0.05$, so $\mathrm{VaR}_{95\%} = -q_{0.05}(R)$. If $q_{0.05}(R) = -2.25\%$, then

$$\boxed{\mathrm{VaR}_{95\%} = 2.25\%}, \qquad \mathbb{P}(R < -2.25\%) \approx 5\%.$$

Equivalently: with 95% probability, the loss does not exceed 2.25% over the horizon.

The Loss Convention

Define the loss variable $\boxed{L = -R}$: large positive L corresponds to a large loss. VaR is then the **upper** quantile of the loss distribution:

$$\boxed{\text{VaR}_{1-\alpha} = q_{1-\alpha}(L) = F_L^{-1}(1-\alpha)}.$$

This gives the identity

$$\boxed{-q_\alpha(R) = q_{1-\alpha}(-R)},$$

which explains why a 95% VaR corresponds equivalently to:

- the 5% **lower-tail** quantile of returns, or
- the 95% **upper-tail** quantile of losses.

Main Methods for Estimating VaR

There are four main approaches to estimating the quantile required for VaR:

1. **Historical** VaR;
2. **Parametric** VaR;
3. **Monte Carlo** VaR;
4. **Conditional** (volatility-model-based) VaR.

1. Historical VaR

No parametric assumption: estimate the lower-tail quantile directly from past returns $R_1, \ldots, R_T$:

$$\boxed{\widehat{\mathrm{VaR}}^{\mathrm{Hist}}_{1-\alpha} = -\widehat{q}_{\alpha}(R)}, \qquad \widehat{\mathrm{VaR}}^{\mathrm{Hist}}_{95\%} = -\widehat{q}_{0.05}(R).$$

Advantage: no Gaussian assumption. Drawback: assumes the historical sample is representative, and tail estimation is imprecise with few extreme observations.

2. Parametric VaR

Assume a distribution for returns, e.g. $\boxed{R \sim \mathcal{N}(\mu, \sigma^2)}$, so $R = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0,1)$. Let $z_\alpha = \Phi^{-1}(\alpha)$. Then

$$\boxed{q_\alpha(R) = \mu + \sigma z_\alpha} \qquad \Longrightarrow \qquad \boxed{\mathrm{VaR}_{1-\alpha}^{\mathrm{Normal}} = -(\mu + \sigma z_\alpha)}.$$

Standard confidence levels

Since $z_{0.05} \approx -1.645$ and $z_{0.01} \approx -2.326$:

$$\boxed{\mathrm{VaR}_{95\%}^{\mathrm{Normal}} = -\mu + 1.645\sigma}, \qquad \boxed{\mathrm{VaR}_{99\%}^{\mathrm{Normal}} = -\mu + 2.326\sigma}.$$

If μ is small relative to σ: $\mathrm{VaR}_{95\%} \approx 1.645\sigma$.

Limitation: financial returns often exhibit skewness and heavy tails, so the Gaussian assumption can underestimate extreme risk.

Deriving the Gaussian Quantile (1/2)

Suppose $\boxed{R \sim \mathcal{N}(\mu, \sigma^2)}$. Standardize: $\boxed{Z = \dfrac{R-\mu}{\sigma}} \sim \mathcal{N}(0,1)$.

By definition, the α-quantile of R satisfies

$$\boxed{\mathbb{P}(R \leq q_\alpha(R)) = \alpha}.$$

Since $\sigma > 0$, subtract μ and divide by σ (inequality direction unchanged):

$$\mathbb{P}\left(\frac{R-\mu}{\sigma} \leq \frac{q_\alpha(R)-\mu}{\sigma}\right) = \alpha \quad \Longrightarrow \quad \mathbb{P}\left(Z \leq \frac{q_\alpha(R)-\mu}{\sigma}\right) = \alpha.$$

Let $\Phi(z) = \mathbb{P}(Z \leq z)$. Therefore

$$\Phi\left(\frac{q_\alpha(R)-\mu}{\sigma}\right) = \alpha.$$

Deriving the Gaussian Quantile (2/2)

Applying Φ^{-1} to both sides:

$$\frac{q_\alpha(R) - \mu}{\sigma} = \Phi^{-1}(\alpha).$$

Define $\boxed{z_\alpha = \Phi^{-1}(\alpha)}$, the α-quantile of the standard normal. Hence

$$\frac{q_\alpha(R) - \mu}{\sigma} = z_\alpha \quad \Longrightarrow \quad q_\alpha(R) - \mu = \sigma z_\alpha.$$

Adding μ:

$$\boxed{q_\alpha(R) = \mu + \sigma z_\alpha}.$$

Beyond the Gaussian: Other Parametric Models and Monte Carlo VaR

Parametric VaR does not require normality: if $R \sim F(\theta)$, then

$$\boxed{\text{VaR}_{1-\alpha} = -F^{-1}(\alpha;\theta)}$$

– e.g. a Student-t distribution for heavier tails.

Monte Carlo VaR

Simulate $R_{t+1}^{(1)}, \ldots, R_{t+1}^{(M)}$ from a chosen model and take $\widehat{\text{VaR}}_{1-\alpha}^{\text{MC}} = -\widehat{q}_\alpha(R^{(1)}, \ldots, R^{(M)})$:

$$\text{Model} \rightarrow \text{Simulate} \rightarrow \text{Empirical distribution} \rightarrow \text{Tail quantile.}$$

Useful when returns depend non-linearly on risk factors (e.g. options).

Conditional VaR Using Volatility Models

Allow the return distribution to change over time:

$$\boxed{R_{t+1} = \mu_{t+1|t} + \sigma_{t+1|t} Z_{t+1}},$$

where $\mu_{t+1|t} = \mathbb{E}_t[R_{t+1}]$ and $\sigma^2_{t+1|t} = \text{Var}_t(R_{t+1})$. If $Z_{t+1} \sim \mathcal{N}(0,1)$,

$$\boxed{\text{VaR}_{1-\alpha,t} = -\left(\mu_{t+1|t} + \sigma_{t+1|t} z_\alpha\right)}, \qquad \text{VaR}_{95\%,t} = -\mu_{t+1|t} + 1.645\,\sigma_{t+1|t}.$$

$\sigma_{t+1|t}$ is estimated by a GARCH model; for GARCH(1,1):

$$\boxed{\sigma^2_{t+1|t} = \omega + \alpha \varepsilon_t^2 + \beta \sigma_t^2}.$$

Key difference vs. unconditional VaR

Unlike a VaR based on constant volatility, the GARCH VaR moves through time: high predicted volatility $\Rightarrow$ higher VaR.

The Conditional Return Model

The predicted VaR combines (i) a GARCH model for time-varying volatility and (ii) a quantile of the **standardized residual** distribution for the tail shape:

$$\boxed{R_{t+1} = \mu_{t+1|t} + \sigma_{t+1|t} Z_{t+1}}, \qquad \boxed{\mathbb{E}[Z_{t+1}] = 0,\ \mathrm{Var}(Z_{t+1}) = 1}.$$

GARCH forecasts $\sigma_{t+1|t}$; e.g. for GARCH(1,1), $\sigma^2_{t+1|t} = \omega + \alpha \varepsilon_t^2 + \beta \sigma_t^2$.

The Conditional Quantile of Returns (1/2)

Let $q_\alpha^Z = F_Z^{-1}(\alpha)$ satisfy $\mathbb{P}(Z_{t+1} \leq q_\alpha^Z) = \alpha$. We seek $q_{\alpha,t}(R_{t+1})$ such that $\mathbb{P}_t(R_{t+1} \leq q_{\alpha,t}(R_{t+1})) = \alpha$.
Substituting the conditional return model:

$$\mathbb{P}_t\left(\mu_{t+1|t} + \sigma_{t+1|t} Z_{t+1} \leq q_{\alpha,t}(R_{t+1})\right) = \alpha.$$

Since $\sigma_{t+1|t} > 0$, rearrange:

$$\mathbb{P}_t\left(Z_{t+1} \leq \frac{q_{\alpha,t}(R_{t+1}) - \mu_{t+1|t}}{\sigma_{t+1|t}}\right) = \alpha \quad \Longrightarrow \quad \frac{q_{\alpha,t}(R_{t+1}) - \mu_{t+1|t}}{\sigma_{t+1|t}} = q_\alpha^Z.$$

The Conditional Quantile of Returns (2/2)

Hence

$$q_{\alpha,t}(R_{t+1}) - \mu_{t+1|t} = \sigma_{t+1|t}\, q_\alpha^Z,$$

and finally, the **general conditional quantile formula**:

$$\boxed{q_{\alpha,t}(R_{t+1}) = \mu_{t+1|t} + \sigma_{t+1|t}\, q_\alpha^Z}.$$

Gaussian GARCH VaR

If $Z_{t+1} \sim \mathcal{N}(0,1)$, then $q_\alpha^Z = \Phi^{-1}(\alpha) = z_\alpha$, so

$$\boxed{q_{\alpha,t}(R_{t+1}) = \mu_{t+1|t} + \sigma_{t+1|t} z_\alpha}.$$

GARCH with an Empirical Standardized-Residual Quantile

Instead of assuming $Z_t \sim \mathcal{N}(0,1)$, estimate its distribution from the GARCH standardized residuals $\boxed{\widehat{z}_t = \widehat{\varepsilon}_t / \widehat{\sigma}_t}$, where $\widehat{\varepsilon}_t = R_t - \widehat{\mu}_t$.
Under a **zero-mean** specification ($\widehat{\mu}_t = 0$): $\widehat{\varepsilon}_t = R_t$, so

$$\boxed{\widehat{z}_t = \frac{R_t}{\widehat{\sigma}_t}}.$$

The empirical α-quantile of the residuals is

$$\boxed{\widehat{q}_\alpha^Z = \widehat{F}_Z^{-1}(\alpha) = \text{Quantile}_\alpha(\widehat{z}_1, \ldots, \widehat{z}_t)}.$$

Unlike the Gaussian case, we do **not** impose $q_\alpha^Z = \Phi^{-1}(\alpha)$: the tail quantile is estimated directly from the historical standardized innovations.

Predicted Return Quantile & a Numerical Example

Combining the GARCH volatility forecast and the empirical residual quantile:

$$\boxed{\widehat{q}_{\alpha,t}(R_{t+1}) = \widehat{\mu}_{t+1|t} + \widehat{\sigma}_{t+1|t}\,\widehat{q}^{Z}_{\alpha}} \overset{\widehat{\mu}=0}{\Longrightarrow} \boxed{\widehat{q}_{\alpha,t}(R_{t+1}) = \widehat{\sigma}_{t+1|t}\,\widehat{q}^{Z}_{\alpha}} = \widehat{\mathrm{VaR}}_{\alpha,t+1|t}.$$

Numerical example

Let $\widehat{\sigma}_{t+1|t} = 2\%$ and $\widehat{q}^{Z}_{0.05} = -1.92$. Then

$$\widehat{q}_{0.05,t}(R_{t+1}) = 0.02 \times (-1.92) = \boxed{-3.84\%}.$$

The estimated conditional 5% return quantile for the next period is -3.84%.

Signed Return Quantile vs. Positive-Loss VaR

Two conventions:

$$\boxed{\mathrm{VaR}^{\mathrm{return}}_{\alpha,t+1|t} = q_{\alpha,t}(R_{t+1})} \quad \text{(typically negative)} \qquad \text{vs.} \qquad \boxed{\mathrm{VaR}^{\mathrm{loss}}_{1-\alpha,t+1|t} = -q_{\alpha,t}(R_{t+1})}.$$

In the previous example: $\mathrm{VaR}^{\mathrm{return}}_{0.05,t+1|t} = -3.84\%$, while $\boxed{\mathrm{VaR}^{\mathrm{loss}}_{95\%,t+1|t} = 3.84\%}$.

Both numbers represent the **same** tail threshold, just with opposite sign conventions.

Gaussian versus Empirical GARCH VaR

Both approaches share the same structure:

$$\boxed{\text{Conditional quantile} = \text{conditional mean} + \text{conditional vol} \times \text{standardized innovation quantile}}.$$

Model	**Innovation quantile**	**Tail shape**
Gaussian GARCH	$q_\alpha^Z = \Phi^{-1}(\alpha)$	parametric
Empirical GARCH	$q_\alpha^Z = \widehat{F}_Z^{-1}(\alpha)$	empirical

Under a zero-mean model, the components combine multiplicatively:

$$\widehat{q}_{\alpha,t}(R_{t+1}) = \underbrace{\widehat{\sigma}_{t+1|t}}_{\text{time-varying scale (GARCH)}} \times \underbrace{\widehat{F}_Z^{-1}(\alpha)}_{\text{tail shape (std. residuals)}}.$$

VaR in Monetary Units & Summary

If V_t is the portfolio value, $\boxed{\text{VaR}^{\text{mon}}_{1-\alpha} = V_t \, \text{VaR}^{R}_{1-\alpha}}$. E.g. $V_t = \text{EUR } 1{,}000{,}000$, $\text{VaR}^{R}_{95\%} = 2.5\% \Rightarrow \text{VaR}^{\text{mon}}_{95\%} = \text{EUR } 25{,}000$.

Method	**Distribution**	**VaR**
Historical	Empirical	$-\widehat{q}_\alpha(R)$
Gaussian parametric	$\mathcal{N}(\mu, \sigma^2)$	$-\mu - \sigma z_\alpha$
General parametric	$F(\theta)$	$-F^{-1}(\alpha; \theta)$
Monte Carlo	Simulated	$-\widehat{q}_\alpha(R^{(1)}, \ldots, R^{(M)})$
Conditional/GARCH	$\mu_{t+1\|t} + \sigma_{t+1\|t} Z_{t+1}$	$-\mu_{t+1\|t} - \sigma_{t+1\|t} z_\alpha$

Setting Up VaR Backtesting

Once a VaR model is estimated, check whether predicted VaR levels are consistent with realized losses. Define the **exception indicator**

$$I_t = \begin{cases} 1 & \text{if a VaR exception occurs at } t \\ 0 & \text{otherwise.} \end{cases}$$

For a 95% VaR, $p = 1 - 0.95 = 0.05$: a correctly specified model should generate $\approx 5\%$ exceptions. Two properties matter:

1. the **frequency** of exceptions should be correct (Kupiec test);
2. exceptions should be **independent over time** (Christoffersen test).

Kupiec Unconditional Coverage (POF) Test

Kupiec (1995): a likelihood-ratio "Proportion of Failures" test. With N forecasts and $x = \sum_{t=1}^{N} I_t$ exceptions, $\widehat{p} = x/N$. Hypotheses:

$$\boxed{H_0 : \mathbb{P}(I_t = 1) = p} \qquad \text{vs.} \qquad \boxed{H_1 : \mathbb{P}(I_t = 1) \neq p}.$$

Likelihood under H_0

$I_t \sim \text{Bernoulli}(p)$, so

$$\boxed{L_0 = (1-p)^{N-x} p^x}.$$

Unrestricted likelihood

MLE of the exception probability is $\widehat{p} = x/N$, giving

$$\boxed{L_1 = \left(1 - \tfrac{x}{N}\right)^{N-x} \left(\tfrac{x}{N}\right)^x}.$$

The Kupiec Likelihood-Ratio Statistic

$$LR_{\text{POF}} = -2\ln\left(\frac{L_0}{L_1}\right) \quad \Longrightarrow \quad \boxed{LR_{\text{POF}} = -2\ln\left[\frac{(1-p)^{N-x}p^x}{\left(1-\frac{x}{N}\right)^{N-x}\left(\frac{x}{N}\right)^x}\right]}.$$

Under H_0: $\boxed{LR_{\text{POF}} \overset{a}{\sim} \chi^2_1}$. We reject the VaR model at level α_{test} if $LR_{\text{POF}} > \chi^2_{1,1-\alpha_{\text{test}}}$, i.e.

$$\boxed{p\text{-value} < \alpha_{\text{test}} \Longrightarrow \text{reject correct unconditional coverage}}.$$

Limitation

The Kupiec test only counts **total** exceptions – it ignores *when* they occur. Two models with the same number of exceptions can differ sharply in whether failures **cluster** in time; clustering suggests the model fails to adapt to changing risk. This motivates the Christoffersen independence test.

The Christoffersen Independence Test: Setup

Christoffersen (1998): test whether exceptions occur independently over time.

$$\boxed{H_0 : I_t \text{ independent of } I_{t-1}} \quad \text{vs.} \quad \boxed{H_1 : I_t \text{ depends on } I_{t-1}}.$$

Transition counts

$$\begin{aligned} n_{00} &= \#\{0 \to 0 \text{ transitions}\} \\ n_{01} &= \#\{0 \to 1 \text{ transitions}\} \\ n_{10} &= \#\{1 \to 0 \text{ transitions}\} \\ n_{11} &= \#\{1 \to 1 \text{ transitions}\} \end{aligned}$$

	$I_t = 0$	$I_t = 1$
$I_{t-1} = 0$	n_{00}	n_{01}
$I_{t-1} = 1$	n_{10}	n_{11}

Conditional Exception Probabilities & Intuition

$$\boxed{\pi_0 = \frac{n_{01}}{n_{00} + n_{01}} = \mathbb{P}(I_t = 1 \mid I_{t-1} = 0)}, \qquad \boxed{\pi_1 = \frac{n_{11}}{n_{10} + n_{11}} = \mathbb{P}(I_t = 1 \mid I_{t-1} = 1)},$$

$$\boxed{\pi = \frac{n_{01} + n_{11}}{n_{00} + n_{01} + n_{10} + n_{11}}}.$$

Intuition

If exceptions are independent, yesterday's state carries no information about today's: $\boxed{\pi_0 = \pi_1 = \pi}$. If $\pi_1 \gg \pi_0$, an exception is far more likely right after another $\Rightarrow$ **clustering of VaR exceptions**.

The Christoffersen Independence Statistic

Likelihood under independence (H_0): $\boxed{L_{\text{ind},0} = (1-\pi)^{n_{00}+n_{10}}\pi^{n_{01}+n_{11}}}$.

Likelihood under the alternative: $\boxed{L_{\text{ind},1} = (1-\pi_0)^{n_{00}}\pi_0^{n_{01}}(1-\pi_1)^{n_{10}}\pi_1^{n_{11}}}$.

The likelihood-ratio statistic:

$$\boxed{LR_{\text{CCI}} = -2\ln\left[\frac{(1-\pi)^{n_{00}+n_{10}}\pi^{n_{01}+n_{11}}}{(1-\pi_0)^{n_{00}}\pi_0^{n_{01}}(1-\pi_1)^{n_{10}}\pi_1^{n_{11}}}\right]} \overset{a}{\sim} \chi_1^2 \text{ under } H_0.$$

$$\boxed{p\text{-value} < \alpha_{\text{test}} \Longrightarrow \text{reject independence of exceptions}}.$$

Conditional Coverage & Summary of the Tests

A good VaR model needs **both** correct coverage and independence. Christoffersen's **Conditional Coverage (CC)** test sums the two LR statistics:

$$LR_{\text{CC}} = LR_{\text{POF}} + LR_{\text{CCI}} \overset{a}{\sim} \chi^2_2, \qquad H_0 : \mathbb{P}(I_t = 1) = p \text{ and } I_t \text{ independent over time}.$$

Test	Question	Statistic	Dist.
Kupiec POF	Correct exception frequency?	LR_{POF}	χ^2_1
Christoffersen IND	Are exceptions independent?	LR_{CCI}	χ^2_1
Christoffersen CC	Both jointly?	$LR_{\text{POF}} + LR_{\text{CCI}}$	χ^2_2

A well-calibrated VaR model generates the correct number of exceptions, **and** those exceptions do not cluster over time.

From Individual VaR to Systemic Risk

VaR measures the tail risk of *one* institution in isolation. In a financial system, we also want to know whether the distress of institution i is associated with distress of the system as a whole. Let $r_{m,t}$, $r_{i,t}$ denote the market and institution-i returns.

Individual VaR

$$\boxed{\mathbb{P}(r_{i,t} \leq VaR_{i,t}(\alpha)) = \alpha}.$$

E.g. $\alpha = 0.05 \Rightarrow VaR_{i,t}(0.05)$ is the 5% lower-tail quantile of i's return (signed-return convention: typically negative).

Definition of CoVaR

CoVaR is the VaR of the system **conditional** on a state $C(r_{i,t})$ of institution i:

$$\boxed{\mathbb{P}\left(r_{m,t} \leq CoVaR_t^{m|C(r_i)}(\alpha) \mid C(r_{i,t})\right) = \alpha}.$$

Institution i in distress

The distress state sets i's return to its own VaR, $r_{i,t} = VaR_{i,t}(\alpha)$:

$$\boxed{\mathbb{P}\left(r_{m,t} \leq CoVaR_t^{m|r_i=VaR_i(\alpha)} \mid r_{i,t} = VaR_{i,t}(\alpha)\right) = \alpha}.$$

Conceptually: $\boxed{VaR_i = \text{tail risk of } i}$, while

$\boxed{CoVaR^{m|i} = \text{tail risk of the market given the state of } i}$.

Why a Benchmark State? ΔCoVaR

A low market CoVaR when i is distressed does not by itself reveal i's *contribution* to systemic risk – the market may already be risky regardless. Compare two states of i:

$$\boxed{\text{Distress: } r_{i,t} = VaR_{i,t}(\alpha) \qquad \text{vs.} \qquad \text{Normal: } r_{i,t} = \text{Median}(r_{i,t})}.$$

ΔCoVaR

$$\boxed{\Delta CoVaR_{i,t}(\alpha) = CoVaR_t^{m|r_i = VaR_i(\alpha)} - CoVaR_t^{m|r_i = \text{Median}(r_i)}}.$$

Under the signed-return convention, a strongly negative $\Delta CoVaR_i$ means the market's lower tail worsens substantially when i moves from normal to distress:

$$\boxed{\text{more negative } \Delta CoVaR_i \Rightarrow \text{larger adverse systemic-risk contribution}}.$$

CoVaR Under a Gaussian Linear Model

To obtain a closed-form ΔCoVaR, assume market and institution returns are jointly Gaussian:

$$\boxed{\begin{pmatrix} R_{m,t+1} \\ R_{i,t+1} \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} \mu_{m,t+1} \\ \mu_{i,t+1} \end{pmatrix}, \begin{pmatrix} \sigma^2_{m,t+1} & \sigma_{mi,t+1} \\ \sigma_{mi,t+1} & \sigma^2_{i,t+1} \end{pmatrix}\right].}$$

Define the conditional correlation $\boxed{\rho_{m,i,t+1} = \dfrac{\sigma_{mi,t+1}}{\sigma_{m,t+1}\sigma_{i,t+1}}}$, i.e.

$\sigma_{mi,t+1} = \rho_{m,i,t+1}\sigma_{m,t+1}\sigma_{i,t+1}$. We want the distribution of $R_{m,t+1}$ conditional on $R_{i,t+1} = r_i$.

The General Conditional-Gaussian Result

For jointly Gaussian $\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix} \right)$, the conditional distribution of X given $Y = y$ is **also Gaussian**:

$$\boxed{X \mid Y = y \;\sim\; \mathcal{N}\left(\mu_X + \frac{\sigma_{XY}}{\sigma_Y^2}(y - \mu_Y),\;\; \sigma_X^2 - \frac{\sigma_{XY}^2}{\sigma_Y^2} \right).}$$

Conditioning shifts **both** the mean and shrinks the variance. We now derive both pieces from scratch.

Deriving the Conditional Mean (1/3)

Because X, Y are jointly Gaussian, decompose X into a part linearly related to Y and an orthogonal residual:

$$\boxed{X = \mu_X + \beta(Y - \mu_Y) + \eta}, \qquad \mathrm{Cov}(\eta, Y) = 0.$$

This is exactly the population linear regression of X on Y. To find β, write $\eta = X - \mu_X - \beta(Y - \mu_Y)$ and impose $\mathrm{Cov}(\eta, Y) = 0$:

$$\mathrm{Cov}\left(X - \mu_X - \beta(Y - \mu_Y),\ Y\right) = 0.$$

Deriving the Conditional Mean (2/3)

By linearity of covariance:

$$\mathrm{Cov}(X, Y) - \beta \,\mathrm{Cov}(Y, Y) = 0.$$

Since $\mathrm{Cov}(X, Y) = \sigma_{XY}$ and $\mathrm{Cov}(Y, Y) = \mathrm{Var}(Y) = \sigma_Y^2$:

$$\sigma_{XY} - \beta\sigma_Y^2 = 0 \qquad \Longrightarrow \qquad \boxed{\beta = \frac{\sigma_{XY}}{\sigma_Y^2}}.$$

The decomposition becomes

$$\boxed{X = \mu_X + \frac{\sigma_{XY}}{\sigma_Y^2}(Y - \mu_Y) + \eta}.$$

Deriving the Conditional Mean (3/3)

Condition on $Y = y$ and take expectations:

$$\mathbb{E}[X \mid Y = y] = \mu_X + \frac{\sigma_{XY}}{\sigma_Y^2}(y - \mu_Y) + \mathbb{E}[\eta \mid Y = y].$$

Since (X, Y) is jointly Gaussian, η and Y are also jointly Gaussian, and we constructed η so that $\mathrm{Cov}(\eta, Y) = 0$. For jointly Gaussian variables, **zero covariance implies independence**, so $\eta \perp Y$ and $\mathbb{E}[\eta \mid Y = y] = \mathbb{E}[\eta] = 0$. Hence

$$\boxed{\mathbb{E}[X \mid Y = y] = \mu_X + \frac{\sigma_{XY}}{\sigma_Y^2}(y - \mu_Y)}.$$

Applying the Result to Market & Institution Returns

Identify $X = R_{m,t+1}$, $Y = R_{i,t+1}$, so $\mu_X = \mu_{m,t+1}$, $\mu_Y = \mu_{i,t+1}$, $\sigma_{XY} = \sigma_{mi,t+1}$. Conditioning on $R_{i,t+1} = r_i$:

$$\boxed{\mu_{m|i} = \mathbb{E}[R_{m,t+1} \mid R_{i,t+1} = r_i] = \mu_{m,t+1} + \frac{\sigma_{mi,t+1}}{\sigma^2_{i,t+1}}(r_i - \mu_{i,t+1})}.$$

Interpretation

$$\mu_{m|i} = \underbrace{\mu_{m,t+1}}_{\text{unconditional market mean}} + \underbrace{\frac{\sigma_{mi,t+1}}{\sigma^2_{i,t+1}}}_{\text{sensitivity to } i} \underbrace{(r_i - \mu_{i,t+1})}_{i\text{'s deviation from its mean}}.$$

$\beta_{m|i}$ as a Regression Coefficient

The coefficient $\boxed{\beta_{m|i} = \dfrac{\sigma_{mi,t+1}}{\sigma^2_{i,t+1}}}$ is exactly the population OLS slope of regressing the market on institution i: $R_{m,t+1} = a + \beta_{m|i} R_{i,t+1} + \eta_{t+1}$, i.e.

$$\boxed{\beta_{m|i} = \frac{\mathrm{Cov}(R_m, R_i)}{\mathrm{Var}(R_i)}} \qquad \Longrightarrow \qquad \boxed{\mu_{m|i} = \mu_m + \beta_{m|i}(r_i - \mu_i)}.$$

- If $r_i = \mu_i$ (institution at its mean): $\mu_{m|i} = \mu_m$ – no shift.
- If $r_i < \mu_i$ and $\sigma_{mi} > 0$: $\mu_{m|i} < \mu_m$ – a negative shock to an institution positively related to the market shifts the market's conditional mean left.

Rewriting the Conditional Mean via Correlation

Recall $\boxed{\rho_{m,i} = \sigma_{mi}/(\sigma_m \sigma_i)}$, so $\sigma_{mi} = \rho_{m,i}\sigma_m\sigma_i$. Substituting into the regression coefficient:

$$\frac{\sigma_{mi}}{\sigma_i^2} = \frac{\rho_{m,i}\sigma_m\sigma_i}{\sigma_i^2} = \boxed{\rho_{m,i}\frac{\sigma_m}{\sigma_i}}.$$

Hence the conditional mean can equivalently be written

$$\boxed{\mu_{m|i} = \mu_m + \rho_{m,i}\frac{\sigma_m}{\sigma_i}(r_i - \mu_i)}.$$

This form makes explicit that the market's response to institution i depends on $\boxed{\text{correlation} \times \text{relative volatility}}$.

Deriving the Conditional Variance (1/2)

The residual is $\eta = X - \mu_X - \dfrac{\sigma_{XY}}{\sigma_Y^2}(Y - \mu_Y)$, so $\mathrm{Var}(\eta) = \mathrm{Var}\left(X - \dfrac{\sigma_{XY}}{\sigma_Y^2} Y\right)$. Using $\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$:

$$\mathrm{Var}(\eta) = \sigma_X^2 + \left(\frac{\sigma_{XY}}{\sigma_Y^2}\right)^2 \sigma_Y^2 - 2\frac{\sigma_{XY}}{\sigma_Y^2}\sigma_{XY} = \sigma_X^2 + \frac{\sigma_{XY}^2}{\sigma_Y^2} - 2\frac{\sigma_{XY}^2}{\sigma_Y^2}.$$

Deriving the Conditional Variance (2/2)

Simplifying:

$$\mathrm{Var}(\eta) = \boxed{\sigma_X^2 - \frac{\sigma_{XY}^2}{\sigma_Y^2}}.$$

Since $\eta \perp Y$ under joint Gaussianity, this is also the conditional variance:

$$\boxed{\mathrm{Var}(X \mid Y = y) = \sigma_X^2 - \frac{\sigma_{XY}^2}{\sigma_Y^2}}.$$

Applied to market/institution returns, using $\sigma_{mi} = \rho_{m,i}\sigma_m\sigma_i$:

$$\sigma_{m|i}^2 = \sigma_m^2 - \frac{\rho_{m,i}^2\sigma_m^2\sigma_i^2}{\sigma_i^2} = \boxed{\sigma_m^2\left(1 - \rho_{m,i}^2\right)}.$$

The Full Conditional-Gaussian Result

Putting the mean and variance together:

$$R_{m,t+1} \mid R_{i,t+1} = r_i \;\sim\; \mathcal{N}\left(\mu_{m,t+1} + \rho_{m,i,t+1}\frac{\sigma_{m,t+1}}{\sigma_{i,t+1}}(r_i - \mu_{i,t+1}),\quad \sigma^2_{m,t+1}\left(1-\rho^2_{m,i,t+1}\right)\right).$$

This is the key mathematical step behind the Gaussian-linear CoVaR and ΔCoVaR formulas: $\mu_{m|i} = \mu_{m,t+1} + \rho_{m,i,t+1}\frac{\sigma_{m,t+1}}{\sigma_{i,t+1}}(r_i - \mu_{i,t+1})$ and $\sigma^2_{m|i} = \sigma^2_{m,t+1}(1-\rho^2_{m,i,t+1})$.

From Conditional Quantile to CoVaR

Since $R_{m,t+1} \mid r_i$ is Gaussian, its α-quantile is $CoVaR_t^{m|r_i}(\alpha) = \mu_{m|i} + z_\alpha \sigma_{m|i}$ with $z_\alpha = \Phi^{-1}(\alpha)$. Substituting:

$$\boxed{CoVaR_t^{m|r_i}(\alpha) = \mu_{m,t+1} + \rho_{m,i,t+1} \frac{\sigma_{m,t+1}}{\sigma_{i,t+1}} (r_i - \mu_{i,t+1}) + z_\alpha \sigma_{m,t+1} \sqrt{1 - \rho_{m,i,t+1}^2}}.$$

Deriving ΔCoVaR (1/2)

Evaluate the market quantile under the **distress** state $r_i = VaR^{\alpha}_{i,t+1}$:

$$CoVaR_t^{m|r_i=VaR_i^{\alpha}} = \mu_{m,t+1} + \rho_{m,i,t+1}\frac{\sigma_{m,t+1}}{\sigma_{i,t+1}}\left(VaR^{\alpha}_{i,t+1} - \mu_{i,t+1}\right) + z_{\alpha}\sigma_{m,t+1}\sqrt{1-\rho^2_{m,i,t+1}},$$

and under the **median** (normal) state $r_i = \text{Median}(R_{i,t+1})$:

$$CoVaR_t^{m|r_i=\text{Med}} = \mu_{m,t+1} + \rho_{m,i,t+1}\frac{\sigma_{m,t+1}}{\sigma_{i,t+1}}\left(\text{Median}(R_{i,t+1}) - \mu_{i,t+1}\right) + z_{\alpha}\sigma_{m,t+1}\sqrt{1-\rho^2_{m,i,t+1}}.$$

Deriving ΔCoVaR (2/2)

Recall $\Delta CoVaR^{\alpha}_{i,t} = CoVaR^{m|r_i=VaR^{\alpha}_i}_t - CoVaR^{m|r_i=\text{Med}}_t$. Subtracting the two expressions, the terms $\mu_{m,t+1}$ and $z_\alpha \sigma_{m,t+1}\sqrt{1-\rho^2_{m,i,t+1}}$ are **identical** under both states and **cancel**; so do the terms in $\mu_{i,t+1}$. We are left with:

$$\boxed{\Delta CoVaR^{\alpha}_{i,t} = \rho_{m,i,t+1}\frac{\sigma_{m,t+1}}{\sigma_{i,t+1}}\left[VaR^{\alpha}_{i,t+1} - \text{Median}(R_{i,t+1})\right]}.$$

ΔCoVaR as a Regression Coefficient

Recall $\boxed{\beta_{m|i,t+1} = \dfrac{\mathrm{Cov}_t(R_{m,t+1}, R_{i,t+1})}{\mathrm{Var}_t(R_{i,t+1})}}$. Writing the covariance via correlation,

$\beta_{m|i,t+1} = \dfrac{\rho_{m,i,t+1}\sigma_{m,t+1}\sigma_{i,t+1}}{\sigma^2_{i,t+1}} = \boxed{\rho_{m,i,t+1}\dfrac{\sigma_{m,t+1}}{\sigma_{i,t+1}}}$. So the ΔCoVaR formula becomes

$$\boxed{\Delta CoVaR^{\alpha}_{i,t} = \beta_{m|i,t+1}\left[VaR^{\alpha}_{i,t+1} - \mathrm{Median}(R_{i,t+1})\right]}$$

$$= \underbrace{\beta_{m|i}}_{\substack{\text{market's sensitivity} \\ \text{to institution } i}} \times \underbrace{(VaR_i - \mathrm{Median}(R_i))}_{\substack{i\text{'s move from} \\ \text{normal to distress}}}.$$

Economic Interpretation of ΔCoVaR

$$\Delta CoVaR^{\alpha}_{i,t} = \rho_{m,i,t+1} \frac{\sigma_{m,t+1}}{\sigma_{i,t+1}} \left[VaR^{\alpha}_{i,t+1} - \text{Median}(R_{i,t+1}) \right]$$

has two components:

- $\boxed{\rho_{m,i,t+1}\sigma_{m,t+1}/\sigma_{i,t+1}}$: how strongly institution i's movements are associated with market movements;
- $\boxed{VaR^{\alpha}_{i,t+1} - \text{Median}(R_{i,t+1})}$: how far i moves going from normal to distress (generally **negative** for a lower-tail VaR).

If $\rho_{m,i,t+1} > 0$ (positively related to the market), then $\Delta CoVaR^{\alpha}_{i,t} < 0$: institution i's distress shifts the market's lower-tail quantile downward.

The Gaussian Special Case

Under a Gaussian distribution, $\boxed{\text{Median}(R_{i,t+1}) = \mu_{i,t+1}}$, and $VaR^{\alpha}_{i,t+1} = \mu_{i,t+1} + \sigma_{i,t+1} z_{\alpha}$, so

$$VaR^{\alpha}_{i,t+1} - \text{Median}(R_{i,t+1}) = \sigma_{i,t+1} z_{\alpha}.$$

Substituting into the ΔCoVaR formula:

$$\Delta CoVaR^{\alpha}_{i,t} = \rho_{m,i,t+1} \frac{\sigma_{m,t+1}}{\sigma_{i,t+1}} (\sigma_{i,t+1} z_{\alpha}) = \boxed{\rho_{m,i,t+1}\, \sigma_{m,t+1}\, z_{\alpha}}.$$

Since $z_{\alpha} < 0$ for lower-tail probabilities ($\alpha = 0.05$), **positive correlation implies a negative** ΔCoVaR.

Summary: VaR, CoVaR, ΔCoVaR

$$\boxed{VaR_i^{\alpha} = \text{tail risk of institution } i},$$

$$\boxed{CoVaR^{m|i,\alpha} = \text{tail risk of the market conditional on the state of } i},$$

$$\boxed{\Delta CoVaR_i^{\alpha} = \text{change in market tail risk when } i \text{ moves from median to distress}}.$$

Under the Gaussian linear assumption:

$$\boxed{\Delta CoVaR_{i,t}^{\alpha} = \beta_{m|i,t+1}\left[VaR_{i,t+1}^{\alpha} - \text{Median}(R_{i,t+1})\right]}, \qquad \boxed{\beta_{m|i,t+1} = \rho_{m,i,t+1}\frac{\sigma_{m,t+1}}{\sigma_{i,t+1}}}.$$

ΔCoVaR measures how much the market's tail-risk threshold changes when institution i moves from a normal state to a distress state.